

ZUOCHENG /JOE/ WANG 王 祚承

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HIGHLIGHTS

Committed **Computer Science** student with proven **leadership** skills, adept at **taking initiative** and **driving projects** to completion. Excellent at **working in teams**, with strong **communication** abilities and a reputation for **quick learning** and **adaptability**.

EDUCATION

University of Pittsburgh, Bachelor of Science September 2023 - est. April 2027
Pittsburgh, PA
Computer Science & Digital Narrative and Interactive Design

- Cumulative GPA: 3.7/4.0
- Honors: **Dean List** (2023 Fall, 2024 Spring)

EXPERIENCES

Anker Innovation - Eufy Security May 2024 - August 2024
Shenzhen, China
Deep Learning Algorithm Developer Internship

- Completed **Deep Learning Specialization** (Coursera) series course and applied experiments using **PyTorch** with **YOLOv5** demo, running on cloud GPUs with **CUDA**.
- Trained and compared **object-detection** models **YOLOv8 Nano** and **YOLOv8 S**, analyzing performance, deployment feasibility on edge devices, conducting **badcase analysis**, and fine-tuning models to ideal results.
- Tested and fine-tuned **CLIP-retrieval** and **LLaVA-1.5** on a test set, optimizing their performance for practical applications.

Lumiere Research December 2022 - March 2023
Researcher

- Analyzed **machine learning** theories and implemented **recommendation systems** with **TensorFlow** using Google COLAB, focusing on movie recommendations.
- Evaluated and compared different recommendation models using the MovieLens dataset to **enhance prediction accuracy and system performance**.
- Addressed **ethical considerations and algorithmic biases**, contributing to the development of fair and transparent recommendation systems.

PROJECTS

SleePurr May 2025 - Present
— An iOS Application Dedicated to Sleep Health, In Progress
Developer, Project Co-founder

- Architected and developed** a full-stack iOS sleep companion app using **SwiftUI**, **HealthKit**, and **Realm**, featuring AI-powered virtual cat interactions and gamified sleep quality improvement through social PK systems.
- Implemented comprehensive data synchronization pipeline** from **HealthKit** to local **Realm** storage and **Supabase** backend, processing sleep segments, heart rate, and HRV data with custom compression algorithms and incremental sync bookmarks.
- Built robust iOS architecture** with dependency injection, unified **OSLog** logging, and coordinator-based navigation.

OreO (<https://sysyishui.itch.io/oreo>)

September 2025

— A 3D Sci-fi Game for Game Jam

Game Designer, Developer

- Led the team in conceptualizing **game design** and mechanics, ensuring strict alignment with Game Jam themes.
- Developed games using **Unity**; spearheaded environment design and implemented core game logic via **C# programming**.
- Leveraged **Generative AI** tools to produce video assets; refined visual content using Adobe Premiere and performed high-fidelity texture post-processing in Photoshop.

Personal Website (<https://zuocheng.me>)

January 2026 - February 2026

Designer, Developer

- Engineered a minimalist, high-performance portfolio using **Next.js 16**, **React 19**, and **Tailwind CSS 4**, utilizing a **data-driven architecture** to dynamically manage and render multi-disciplinary content across software, photography, and music.
- Engineered a "scroll-to-color" UI using the Intersection **Observer API** for immersive, real-time grayscale-to-color transitions.
- Implemented **advanced UI patterns**, including responsive masonry layouts for photography collections and cinematic video integrations with custom **CSS** grain filters, ensuring a polished and performant experience across all device sizes.

Best Neighborhood in Pittsburgh (Intenal Project)

November 2023 - December 2023

Developer

- Innovatively designed and used coordinates from a database to create coordinate maps to objectively sort Pittsburgh's 100+ neighborhoods by visualizing crime rate and emergency data on a map.
- Using **JupyterLab** + GitHub, collaborated effectively within a team of three, contributing to the strategic planning of the overall structure of the project. Organized the presentation and illustrated our result.
- Demonstrated expertise in **Python** and **Pandas** throughout the development process, ensuring accuracy and reliability in the creation of the neighborhood ranking system.

SKILLS

- **Programming:** JAVA, Python, Swift, C, C++, HTML
- **Productivity:** Microsoft Word, Microsoft PowerPoint, Microsoft Excel, Canva, iA Presenter, etc.,
- **Creativity:** Adobe Photoshop, Adobe Premiere, Adobe Lightroom, FL Studio, Unity, Unreal Engine, etc.,

ORGANIZATION/LEADERSHIP

CSSA (Chinese Students and Scholars Association) at University of Pittsburgh

Department of Media

November 2025 - est. August 2025

ScrumAlliance

Certified ScrumMaster

2026

TG 13th Senior Highschool Students Council

Vice President

August 2021 - July 2022